Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2022-P1-Model)** **Date: 19-04-2026**

Time: 09.00Am to 12.00Pm **GTA-25** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	159	**2**	19.96 to 19.99	**3**	35.84 to 35.92	**4**	0.16 to 0.17	**5**	44	**6**	1.30 – 1.31
7	6	**8**	32	**9**	ABCD	**10**	ABD	**11**	AD	**12**	ABD
13	ABD	**14**	AC	**15**	C	**16**	B	**17**	D	**18**	B

PHYSICS

19	0	**20**	684	**21**	0.1	**22**	4	**23**	6.75	**24**	59
25	2	**26**	0.13	**27**	AB	**28**	ABC	**29**	ACD	**30**	ACD
31	ACD	**32**	ABCD	**33**	D	**34**	B	**35**	A	**36**	D

CHEMISTRY

37	0	**38**	13	**39**	0.62 to 0.63	**40**	4	**41**	7	**42**	228
43	3	**44**	13	**45**	ABC	**46**	BC	**47**	ABD	**48**	ACD
49	ABD	**50**	B	**51**	C	**52**	B	**53**	D	**54**	C

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SOLUTIONS
MATHEMATICS

1. $P=159$

Periodicity of $\sin Sx$ is $\frac{2}{5}$...

Total 80 simoscridal Waves in $x\in[0,32]$. There will be 2 points of intersections of $\log_2 x=5\sin(5\pi x)$ in each wave exelts one wave from $x=\frac{4}{5}$ to $x=\frac{6}{5}$.

$\therefore$ Total number of solution is $2\times80-1=159$.

Also for Q

$$\left(\sin x^2y+2x^2y\cos x^2y\right)dx+x^3\cos\left(x^2y\right)dy=-dy$$

$$d\left[x\sin\left(x^2y\right)\right]=-dy$$

$$x\sin\left(x^2y\right)+y=c$$

$$y(0)=1\Rightarrow c=1$$

$$\therefore x\sin\left(x^2y\right)+y=1$$

$$y(\alpha)=1$$

$$\alpha\sin\left(\alpha^2\right)=0$$
$$\therefore\theta=0$$

2. $\cos3A+\cos3B+\cos3C=1$

$$2\cos\left(\frac{3A+3B}{2}\right)\cos\left(\frac{3A-3B}{2}\right)=2\sin^2\frac{3C}{2}$$

$$-2\sin\frac{3C}{2}\cos\left(\frac{3A-3B}{2}\right)=2\sin^2\frac{3C}{2}$$

$$\sin\frac{3C}{2}=0$$

$\therefore\ C=120^\circ$

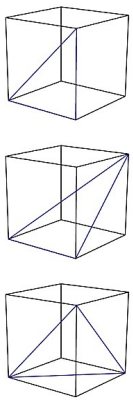
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$$\cos 120 = \frac{10^2 + 13^2 - x^2}{2 \times 10 \times 13}$$

$$x^2 = 399$$

$$x = \sqrt{399}$$

3.



Since there are 8 vertices of a cube, there are $\binom{8}{3} = 56$ total triangles to consider. They fall into three categories: there are those which are entirely contained within a single face of the cube (whose sides are two edges and one face diagonal), those which lie in a plane perpendicular to one face of the cube (whose sides are one edge, one face diagonal and one space diagonal of the cube) and those which lie in a plane oblique to the edges of the cube, whose sides are three face diagonals of the cube.

Each face of the cube contains $\binom{4}{3} = 4$ triangles of the first type, and there are 6 faces, so there are 24 triangles of the first type. Each of these is a right triangle with legs of length 1, so each triangle of the first type has area $\frac{1}{2}$.

Each edge of the cube is a side of exactly 2 of the triangles of the second type, and there are 12 edges, so there are 24 triangles of the second type. Each of these is a right triangle with legs of length 1 and $\sqrt{2}$, so each triangle of the second type has area $\frac{\sqrt{2}}{2}$.

Each vertex of the cube is associated with exactly one triangle of the third type (whose vertices are its three neighbors), and there are 8 vertices of the cube, so there are 8 triangles of the third type. Each of the these is an equilateral triangle with sides of length $\sqrt{2}$, so each triangle of the third type has area $\frac{\sqrt{3}}{2}$.

Thus the total area of all these triangles is

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$24\cdot\frac{1}{2}+24\cdot\frac{\sqrt{2}}{2}+8\cdot\frac{\sqrt{3}}{2}=12+12\sqrt{2}+4\sqrt{3}=12+\sqrt{288}+\sqrt{48}$ and the answer is

$$12+288+48=348$$

4. $z^{1997}=1=1(\cos 0+i\sin 0)$

Define $\theta=2\pi/1997$. By De Moivre's Theorem the roots are given by

$$z=\cos(k\theta)+i\sin(k\theta),\ k\in\{0,1,\ldots,1996\}$$

Now, let v be the root corresponding to $m\theta=2m\pi/1997$, and let w be the root corresponding to $n\theta=2n\pi/1997$. Then

$$\begin{aligned}|v+w|^2 &=(\cos(m\theta)+\cos(n\theta))^2+(\sin(m\theta)+\sin(n\theta))^2\\ &=2+2\cos(m\theta)\cos(n\theta)+2\sin(m\theta)\sin(n\theta)\end{aligned}$$

The cosine difference identity simplifies that to

$$|v+w|^2=2+2\cos((m-n)\theta)$$

We need $|v+w|^2\geq 2+\sqrt{3}$, which simplifies to

$$\cos((m-n)\theta)\geq\frac{\sqrt{3}}{2}$$

Hence $-\frac{\pi}{6}\leq(m-n)\theta\leq\frac{\pi}{6}$, or alternatively, $|m-n|\theta\leq\frac{\pi}{6}$.

Recall that $\theta=\frac{2\pi}{1997}$. Thus,

$$|m-n|\leq\frac{\pi}{6}\cdot\frac{1997}{2\pi}=\frac{1997}{12}=166.$$

Therefore, m and n cannot be more than 166 away from each other. This means that for a given value of m, there are 332 values for n that satisfy the inequality; 166 of them $>m$, and 166 of them $<m$ ($m\neq n$ since the problem says that m and n are distinct). Since m and n must be distinct, n can have 1996 total possible values for each value of m. Therefore, the probability is $\frac{332}{1996}=\frac{83}{499}$.

5. Using Simon's Favorite Factoring Trick, we rewrite as

$$[(x+7)(y-3)]^n=(x+7)^n(y-3)^n.$$

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Both binomial expansions will contain $n+1$ non-like terms; their product will contain $(n+1)^2$ terms, as each term will have an unique power of x or y and so none of the terms will need to be collected.

Hence $(n+1)^2 \geq 1996$, the smallest square after 1996 is $2025 = 45^2$, so our answer is $45-1=044$.

6. Let the sum of all of the terms in the sequence be $\mathbb{S}$. Then for each integer k, $x_k = \mathbb{S} - x_k - k \Rightarrow \mathbb{S} - 2x_k = k$. Summing this up for all k from $1,2,\ldots,100$,

$$100\mathbb{S} - 2\left(x_1 + x_2 + \cdots + x_{100}\right) = 1 + 2 + \cdots + 100$$

$$100\mathbb{S} - 2\mathbb{S} = \frac{100 \cdot 101}{2} = 5050$$

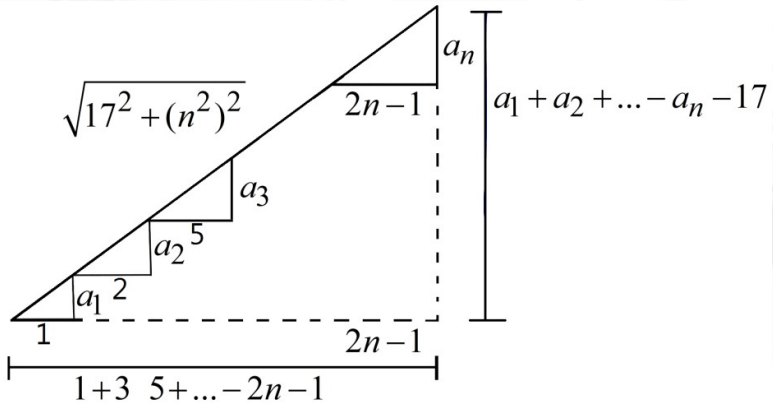
$$\mathbb{S} = \frac{2525}{49}$$

Now, substituting for x_{50}, we get $2x_{50} = \frac{2525}{49} - 50 = \frac{75}{49} \Rightarrow x_{50} = \frac{75}{98}$, and the answer is $75+98=173$.

7. Let $A = \sum_{k=1}^{n} \sqrt{(2k-1)^2 + a_k^{\,2}}$

$$\therefore \quad A = \sqrt{1+a_1^2} + \sqrt{3^2+a_2^2} + \sqrt{5^2+a_3^2} + \ldots \sqrt{(2n-1)^2+a_n^2}$$

For $S_n \Rightarrow$



$$\therefore \ S_n = \sqrt{289+n^4}$$

$$S_n^2 - (n^2)^2 = 289$$

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Solving, we get

$\boxed{n=12}$

8. $f(x)=f(x+6)$

$$\therefore \lim_{x\to 0}\frac{f^2(x)-2f(x)f(0)+f^2(0)}{x\left[\tan^{-1}\left(\frac{x}{2-x}\right)\right]}$$

$$\lim_{x\to 0}\left[\frac{f(x)-f(10)}{x}\right]^2(2-x)$$

$$2(f'(0))^2=32$$

9. Let $a-b=p, b-c=q, c-a=r$

$$\therefore\ p^7+q^7+r^7=0$$

$$\therefore\ p+q+r=0$$

Let $r_2=pq+qr+rp$

$r_3=pqr$

Consider a cubic whose roots are p,q,r is

$$x^3+r_2x-r_3=0$$

$$x^3=-r_2x+r_3$$

$$x^4=-r_2x^2+r_3x$$

$$x^5=r_2^2x-r_2r_3+r_3x^2$$

$$x^6=r_2^2x^2-2r_2r_3x+r_3^2$$

$$x^7=\left(-r_2^3+r_3^2\right)x-2r_2r_3x^2+r_2^2r_3$$

$$p^7+q^7+r^7=\left(-r_2^3+r_3^2\right)(p+q+r)-2r_2r_3\left(p^2+q^2+r^2\right)+p^1+q^1+r^7=Sr_2^2r_3=0$$

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$$p^7+q^7+r^7=Sr_2^2r_3=0$$

$$(pq+qr+rp)^2(pqr)=0$$

None a,b,c are distant

$\therefore\ p, q, r\neq 0$

$\therefore\ pq+qr+rp=0$

$$(a-b)(b-c)+(b-c)(c-a)+(c-a)(a-b)=0$$

$$a^2+b^2+c^2=ab+bc+ca$$

(B) $p,q,r\neq 0,\ \dfrac{1}{p^2}+\dfrac{1}{q^2}+\dfrac{1}{r^2}=\dfrac{p^2q^2+q^2r^2+p^2+r^2}{p^2q^2r^2}=0$

(C) As $pq+qr+rp=0$

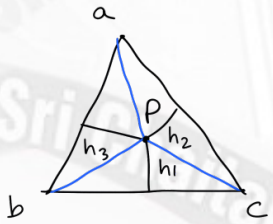
$x^3-pqr=0\Rightarrow$ for $x=p, q, r$

$x^3=pqr$

$\therefore\ p^3=q^3=r^3$

$p^3=q^3=r^3\Rightarrow|p|=|q|=|r|$

$\Rightarrow|a-b|=|b-c|=|c-a|$



Area of $\Delta ABC=\dfrac{\sqrt{3}}{4}|a-b|^2=\dfrac{1}{2}|a-b|(h_1+h_2+h_3)$

None $\dfrac{c-a}{b-a}=e^{i\frac{\pi}{3}}=\dfrac{\sqrt{3}}{2}$

10. $g(x)=\ln(1+x)\forall\ x>0$

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$$\lim_{n\to\infty}\left(1+\frac{1+\frac{1}{2}+\ldots\ldots+\frac{1}{n}}{n^2}\right)^n$$

limit is in the form of 1^∞

$$\beta=\exp\left(\lim_{n\to\infty}\frac{1+\frac{1}{2}+\frac{1}{3}+\ldots\ldots+\frac{1}{n}}{n^2}\right)$$

$$0\le 1+\frac{1}{2}+\frac{1}{3}+\ldots\ldots+\frac{1}{n}\le 1+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\ldots..+\frac{1}{\sqrt{n}}\le 2\sqrt{n}-1$$

Taking limit $(n\to\infty)$

$\beta\ \boldsymbol{b}=\exp(0)$ (from sandwich)

$\beta=1 \qquad \therefore \alpha=e^{1/2}$

11. $f(x)=e^x(2x-1)$

$f(x)$ has local minimum at $x=-1/2$

As $x\to 0^+ f(x)\to -1$ and RHS $\to 1\therefore$ B is wrong

use LMVT

12. $A^T=A-B\Rightarrow A=A^T-B^T$

$B^T=B-C$

$A=A^T-B^T=(A-B)-(B-C)=A-2B+C$

$\Rightarrow C=2B=|C|=2^n|B|$

$B^T=B-C=B-2B=-B$

$\Rightarrow B^T=-B=|B|=0$

$|A+B|=|A^T+B^T|=|(A-B)+(B-C)|$

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$= |A - C| = |A - 2B|$

$\Rightarrow |A + B| = |A - 2B|$

13. Total number of determinants formed will be $12^9 = N$.
Value of determinants will be equal in magnitude and opposite in sign for the same number of times.
Hence sum of all det. is 0.

14. Let $P(h,k)$

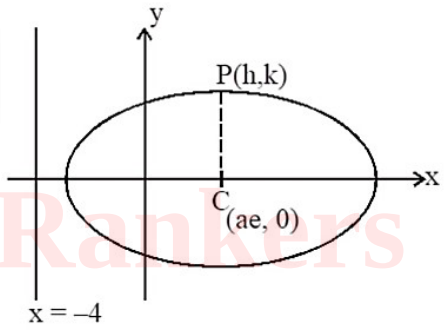
$\Rightarrow h = ae$ and $k = b$

Also $\frac{a}{e} - ae = 4$

$\Rightarrow a\left(1 - e^2\right) = 4e$

$\Rightarrow a^2\left(1 - e^2\right) = 4ae$

$\Rightarrow b^2 = 4ae$



$\Rightarrow k^2 = 4h \Rightarrow$ locus is $y^2 = 4x$

$\Rightarrow S_1 \equiv y^2 - 4x = 0$ & $L_1 \equiv x + 1 = 0$ (directrix)

$A(-1,0)$

Normal to $y^2 = 4x$ is $y = mx - 2m - m^3$

$\Rightarrow 0 = -m - 2m - m^3$

$\Rightarrow m = 0$

15. L_1 and L_2 are coplanar

So, they are intersecting $= 6 - \lambda = 1 + \mu$ and $-1 + 2\lambda = -1 + 3\mu\lambda = 3$ and $\mu = 2$

$$\cos\theta = \frac{\left(2\vec{b} - \vec{a}\right)\cdot\left(\vec{a} + 2\vec{b}\right)}{\left|2\vec{b} - \vec{a}\right|\left|\vec{a} + 3\vec{b}\right|} = \frac{1}{\sqrt{2}}$$

$\theta = \frac{\pi}{4}$ so, $[\tan\theta] + 3 = 4$

Point of intersection is $3\vec{a} + 5\vec{b}$.

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It is distance from origin is $\sqrt{34}=d$ So, $[d]=5$

16. (A) $f(x)=x^3-3x+a, f'(x)=3x^2-3$

$\therefore f'(x)=0$ at $x=-1$ and $x=1$

at $x=-1, f(x)$ has maxima,

at $x=1, f(x)$ has minima.

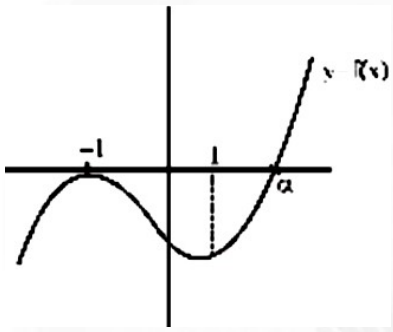
(A) If $f(x)$ has only one real solution then $f(-1)\cdot f(1)>0$

$\Rightarrow (a+2)(a-2)>0$

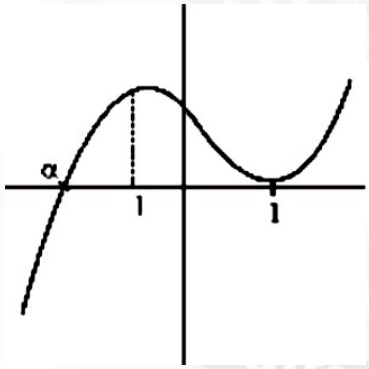
$\therefore a\in(-\infty,-2)\cup(2,\infty)$.

if $a<-2$

as $\alpha<-1, f(x)=\alpha$ has only one real solution.



We have to find no. of solutions for $f(x)=\alpha$ as $\alpha>1$, we get only one real solutions.



(B) $f(x)=0$ has exactly two real solutions

$\therefore f(1)\times f(-1)=0$

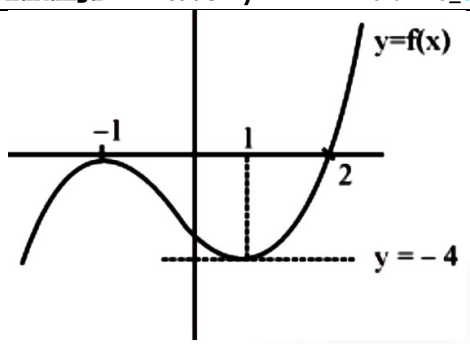
$\therefore a=-2$ or $a=2$

If $a=-2$

$\Rightarrow f(x)=x^3-3x-2=0$

roots are $x=-1,-1,2$

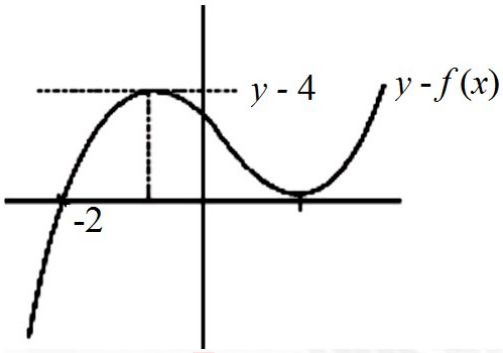
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total no. of solutions for $f(x)=-1$

$f(x)=2$ are four.

If $a=2$



$f(x)=0$ has roots $1,1,-2$

$f(x)=1, f(x)=-2$ has total four solutions.

(D) $\alpha\in(-1,0)\Rightarrow f(x)=\alpha$ has 3 solutions

$\beta\in(1,2)\Rightarrow f(x)=\beta$ has 1 solution

$\gamma\in(-2,-1)\Rightarrow f(x)=\gamma$ has 3 solutions

17. (P) $\int_0^1\left(f^4(x)-8x^2\cdot f^2(x)+16x^4\right)dx$

$\Rightarrow\int_0^1\left(f^2(x)-4x^2\right)^2=0\Rightarrow f^2(x)=4x^2$

(Q) $f''(x)=-\dfrac{4x^3}{2\sqrt{1-x^4}}\Rightarrow f''(x)=0$ at $x=0$ and $f'(x)$ changes sign around it

(R) $g'(x)=x^2\left(\int_1^x f(u)du\right)+f(x)\cdot\left(\int_0^x t^2dt\right)$

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$$g'(1)=f(1)\int_0^1 t^2dt=\frac{f(1)}{3}=1$$

$$g''(x)=2x\int_1^x f(u)du+x^2f(x)+f(x)\cdot x^2+f'(x)\int_0^x t^2dt$$

$$g''(1)=2f(1)+\frac{f'(1)}{3}=6+\frac{f'(1)}{3}$$

$$\Rightarrow g'(1)+g''(1)=7+\frac{f'(1)}{3}$$

$$I_2=-\frac{1}{3}\int_0^{-1}\frac{dt}{e^{-t}(2+t)}\left(\text{ put }-x^3=t\right)$$

$$=-\frac{1}{3}\int_0^{-1}\frac{e^t}{(1+1+t)}dt$$

Put $1+t=z$

$$=\frac{1}{3}\int_0^1\frac{e^{z-1}}{1+z}dz=\frac{1}{3e}I_1$$

18. Sol.

$$(x-2010)P(x+67)=xP(x)\forall x\in R$$

$$(x-30\cdot 67)P(x+67)=xP(x)$$

$$\text{at } x=0\rightarrow P(67)=0\rightarrow x=67 \text{ is a root}$$

$$\text{at } x=67\rightarrow P(2\cdot 67)=0\rightarrow x=2\cdot 67 \text{ is a root}$$

similarly

$$\text{at } x=30\cdot 67\rightarrow P(30\cdot 67)=0\rightarrow x=30\cdot 67 \text{ is a root}$$

$$P(x)=(x-67)(x-2\cdot 67)\ldots(x-30\cdot 67)Q(x)$$

Where $Q(x)$ is an unknown polynomial

$$P(x+67)=x(x-67)\ldots(x-29\cdot 67)Q(x+67)$$

$$x(x-67)\ldots(x-30.67)Q(x+67)=x(x-67)\ldots(x-30\cdot 67)Q(x)$$

$\forall x\in R$

$Q(x)=Q(x+67)\forall x\in R$

$Q(x)=K$ (some non-zero arbitrary constant)

$$P(x)=K(x-67)(x-2\cdot 67)\ldots(x-30\cdot 67)$$

$$\therefore \alpha_1=67,\alpha_2=2\cdot 67,\ldots\alpha_n=30\cdot 67$$

$$n=30,\ B=\frac{67(1+2+\ldots+30)}{30\cdot 67}=\frac{31}{2}$$

$$\frac{B}{2}=7\cdot 75,C=\frac{67\cdot 67}{15\cdot 67\cdot 67}=2,\frac{\alpha_n}{10\alpha_1}=3$$

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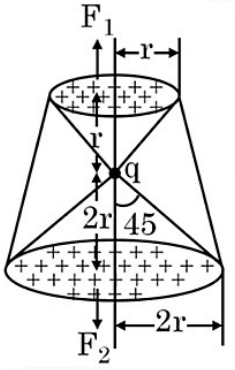
19. Element of charge $dQ = \sigma dA$

E = electric field by q at dA

Force : $F = \int \sigma dAE \cos\theta$ on disc $= \int \vec{E}.d\vec{A} = \sigma(flux)$

$= \sigma \frac{q}{2\varepsilon_0}(1-\cos\alpha)\downarrow$

In our problem



$F_1 = \frac{\sigma q}{2\varepsilon_0}(1-\cos 45^\circ)$

$F_2 = \frac{\sigma q}{2\varepsilon_0}(1-\cos 45^\circ)$

$F_1 = F_2$

F_{net} by frustum on $q = 0$

20. To get minima after maxima $\Delta x = \frac{\lambda}{2}$

On displacing arm by x, $\Delta x = 2x$

Equate $2x = \frac{\lambda}{2} \Rightarrow \lambda = 4x = 4 \times 9.5\,\text{cm} = 38\,\text{cm}$

$u = f\lambda = 1800 \times \frac{38}{100} = \boxed{684\,\text{m/s}}$

21. $\frac{dU}{dt} = \frac{1}{2}k \cdot 2x \cdot \frac{dx}{dt}$

$= kx \cdot v_{rel}$

$= F \times V_{rel}$

$20 = F \times 2$

$\Rightarrow F = 10\text{N}$

22. $W_{net} = \text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2}(S_0)(T_0) = \frac{1}{2} S_0 T_0$

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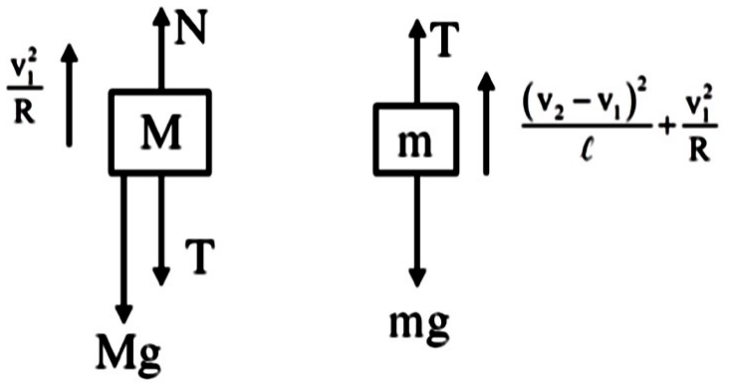
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$$Z=\frac{|Q_B \to C|}{W_{net}}=\frac{2}{\frac{1}{2}}\frac{S_0\ T_0}{S_0\ T_0}$$

$$Z=\frac{1}{1/2}=4$$

23. $W_{ext}=\Delta U_{Water}+\Delta U_{Cube}$

24. Sol.



$$N-T-Mg=M\frac{v_1^2}{R}$$

$$T-mg=\left(\frac{(v_2-v_1)^2}{\ell}+\frac{v_1^2}{R}\right)m$$

25. $C=C_v+\frac{PdV}{ndT}$

$$C=C_v+\frac{RT}{V}\frac{dV}{dT}$$

$$C=\frac{3R}{2}+\frac{RT_0}{V_0}\left(-\frac{V_0}{T_0}\right)$$

$$C=\frac{3R}{2}-R=\frac{R}{2}$$

26. **KVL:** $-\left(L_1\frac{di}{dt}+NS\frac{dB}{dt}\right)-\left(L_2\frac{di}{dt}\right)=0$

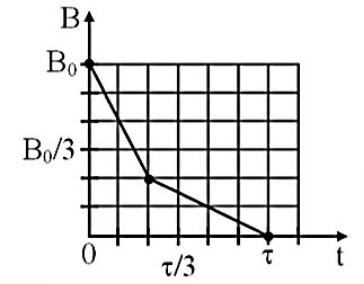
$$(L_1+L_2)\frac{di}{dt}=-NS\frac{dB}{dt}$$

$$(L_1+L_2)\int_0^i di=-NS\int_{B_0}^{B_0} dB$$

$$i=\frac{-NS}{L_1+L_2}(B-B_0)$$

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$$q = \int i dt = \frac{-NS}{L_1 + L_2}\left[\int_0^{\tau} B dt - \int_0^{\tau} B_0 dt\right]$$



$\int_0^{\tau} Bdt$ = Area under curve

$$= \frac{1}{2}\left(\frac{4\ B_0}{3}\right)\frac{\tau}{3} + \frac{1}{2}\frac{2\tau}{3}\frac{B_0}{3} = \frac{B_0\tau}{3}$$

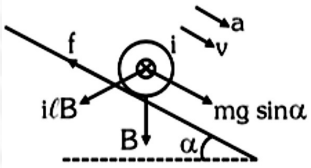
$$\text{So, } q = \frac{-NS}{L_1 + L_2}\left[\frac{B_0\tau}{3} - B_0\tau\right]$$

$$= \frac{NS}{(L_1 + L_2)}\left[\frac{2\ B_0\tau}{3}\right] = \frac{2NSB_0\tau}{15\ L}$$

27. Initially no. emf no. current.

$$\text{So, } a = \frac{g\sin\alpha}{1 + (I/MR^2)} = \frac{g\sin\alpha}{2}$$

$$\text{At any time t: } i = \frac{(v\cos\alpha)B\ell}{R}$$



Side view:

$$\Sigma\tau_{ICR} = I\alpha$$

$$(mg\sin\alpha - i\ell B\cos\alpha)r = \left(2mr^2\right)\frac{a}{r}$$

$$mg\sin\alpha - \frac{vB^2\ell^2\cos^2\alpha}{R} = 2\ m\frac{dv}{dt}$$

$$\int_0^t \frac{dt}{2\ m} = \int_0^v \frac{dv}{k_1 - k_2 v}$$

$$\frac{\left[\ell n(K_1 - K_2 v)\right]_0^v}{-K_2} = \frac{t}{2\ m}$$

$$\ell n\left(1 - \frac{K_2}{K_1}v\right) = \frac{-K_2}{2\ m}t$$

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$$v=\frac{K_1}{K_2}\left(1-e^{-\frac{K_2}{2n}t}\right)$$

So, $\tau=\frac{2\,m}{K_2}=\frac{2mR}{B^2cos^2\alpha\ell^2}$

28. Since apparent frequency is constant for some time interval, hence their velocity along line of joining are constant.

29. Force on q due to dipole

$$F=\frac{1}{4\pi\varepsilon_0}\frac{2pq}{(x+y+z)^3}$$

$$V_A=V_0$$

$$(V_A)_p+(V_A)_{\text{induced charges}}=\frac{-1}{4\pi\varepsilon_0}\frac{p}{(a+y)^2}$$

$$(V_A)_{\text{induced charges}}=\frac{1}{4\pi\varepsilon_0}\frac{p}{y^2}-\frac{1}{4\pi\varepsilon_0}\frac{p}{(x+y)^2}$$

$$V_A=-\frac{p}{4\pi\varepsilon_0(x+y)^2}$$

30. $\Delta\phi_{1,4}=\frac{2\pi}{\lambda}(6\mu\ell)+\pi\Rightarrow\frac{2\pi}{\lambda}(6\mu\ell)+\pi=2n\pi$ for constructive $\Rightarrow 12\mu\ell=(2n-1)\lambda$

$\Delta\phi_{2,3}=\frac{2\pi}{\lambda}(2\mu\ell)\Rightarrow\frac{2\pi}{\lambda}(2\mu\ell)=2n\pi$ for constructive $\Rightarrow 2\mu\ell=n\lambda$

$\Delta\phi_{1,3}=\frac{2\pi}{\lambda}(4\mu\ell)+\pi\Rightarrow\frac{2\pi}{\lambda}(4\mu\ell)+\pi=(2n+1)\pi$ for destructive $\Rightarrow 4\mu\ell=n\lambda$

$\Delta\phi_{2,4}=\frac{2\pi}{\lambda}(4\mu\ell)\Rightarrow\frac{2\pi}{\lambda}(4\mu\ell)=(2n+1)\pi$ for destructive $\Rightarrow 8\mu\ell=(2n+1)\lambda$

31. $x^2+h^2=\ell^2$

$$x\cdot\frac{dx}{dt}=\ell\cdot\frac{d\ell}{dt}\Rightarrow xv=\ell u$$

$$v=\frac{\ell}{x}u\Rightarrow v=u\sec\alpha$$

$$xa+v^2=u^2$$

$$a=\frac{u^2-v^2}{x}=-\frac{u^2\tan^2\alpha}{h\cot\alpha}=-\frac{v^2\sin^2\alpha}{h\cot\alpha}$$

$-\alpha$ is increasing & u is constant as a is increasing

$T\cos\alpha=ma$

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$$\Rightarrow T = \frac{mv^2\sin^3\alpha}{h\cos^2\alpha}$$

Block is moving on a straight line, so there is no normal acceleration.

32. Loss of heat by water = Heat required to convert water into vapour

$$-(m_0 - \mu t)sdT = L_v(\mu dt)$$

$$-\int_{T_0}^{T} dT = \frac{\mu L_v}{S}\int_0^t \frac{dt}{m_0 - \mu t}$$

$$(T_0 - T) = -\frac{L_v}{s}\ell n\left(\frac{m_0 - \mu t}{m_0}\right)$$

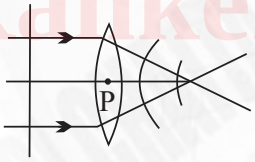
$$T = T_0 + \frac{L_v}{s}\ell n\left(\frac{m_0 - \mu t}{m_0}\right)$$

$$t = \frac{m_0}{\mu}\left[1 - e^{-\frac{s}{L_v}(T_0 - T)}\right]$$

33. Sol.

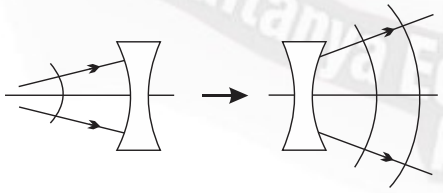
(P)  (5)

(Reflected)

(Q)  (3)

(Reflected)

(R)  (4)

(S)  (1)

34. Unpolarised light will turn to plane polarised with an intensity of I/2 after passing through the first polaroid. After this there are $(n+1)$ polaroids at a pass-axis angle of

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$\frac{\pi}{(n+1)}$ with each other, each will rotate the plane of polarisation by this angle and the intensity drops to a $\cos^2\left(\frac{\pi}{(n+1)}\right)$ factor.

Therefore, $I_{final} = \frac{I}{2}\left[\cos^2\left(\frac{\pi}{(n+1)}\right)\right]^{(n+1)}$

35. $a_1 =$ acceleration of m w.r.t to T

$a_2 =$ = acceleration of 2m w.r.t to T

$a =$ acceleration of 'T' w.r.t to ground

$(\sum f_x) = 0$

$= m(a+a_1) + 2m(a-a_2) + (m+5m+2m)a = 0$

$a + a_1 + 2a - 2a_2 + 8a = 0$

$11a + a_1 - 2a_2 = 0$

For $m, mg - ma = 2ma_1$

$a_1 = \frac{g-a}{2}$

For $2m, 2mg + 2ma = 4ma_2$

$a_2 = \frac{g+a}{2}$

$11a + \frac{g-a}{2} - 2\left(\frac{g+a}{2}\right) = 0$

$11a + \frac{g}{2} - \frac{a}{2} - g - a = 0$

$19a = g$

$a = \frac{g}{19}$

$a_1 = \frac{g-a}{2} = \frac{g-(g/19)}{2} = \frac{9g}{19}$

$a_2 = \frac{g+a}{2} = \frac{10g}{19}$

$a_{12} = a_1 + a_2 = g$

36. $\frac{Mg}{S} + \rho gH = \frac{1}{2}\rho v_B^2$

$\frac{Mg}{S} + \rho gH + 2\rho gH = \frac{1}{2}2\rho v_A^2$

$F = \rho\frac{S}{10}v^2$

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CHEMISTRY

37. $_{80}Hg: 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6, 5s^2, 4d^{10}, 5p^6, 6s^2, 4f^{14}, 5d^{10}$

$w = 2+6+10 = 18$

$\left(3s^2, 3p^6, 3d^{10}\right)$

$x = 10+10+10 = 30$

$\left(3d^{10}, 4d^{10}, 5d^{10}\right)$

$y = 8$

$(2p_z, 3p_z, 4p_z, 5p_z$ each containing 2 electrons).

$z = 40$

[Hg has 80 electrons, and it has a closed shell configuration, exactly half of the electrons will have $m_s = +\frac{1}{2}$ & half will have $m_s = -\frac{1}{2}$]

$(w+x)-(y+z)$

$=(18+30)-(8+40)$

$=0$

38. The titrant is 0.1 M NaOH being added.

at $25^\circ C, [NaOH] = [OH^-] = 0.1M$

so,

$pOH = 1$

$pH = 14-1$

$= 13$

The volume of acid or the equivalence point doesn't affect the pH of titrant itself.

39. C_6H_{10}

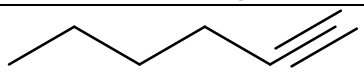
$DU = \frac{14-10}{2} = 2$

y = alkynes that can show positive Tollens' test = terminal alkynes

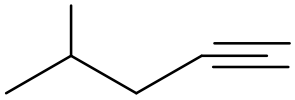
x = total number of alkynes

terminal alkynes for C_6H_{10}

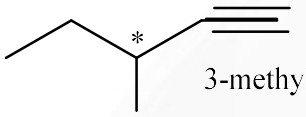
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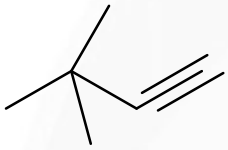
hex-1-yne



4-methyl pent-1-yne



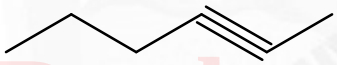
3-methyl pent-1-yne (d & l)



3, 3 dimethyl but-1-yne

$\therefore \quad y = 5$

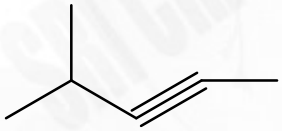
Other isomers (non-terminal)



hex-2-yne



hex-3-yne

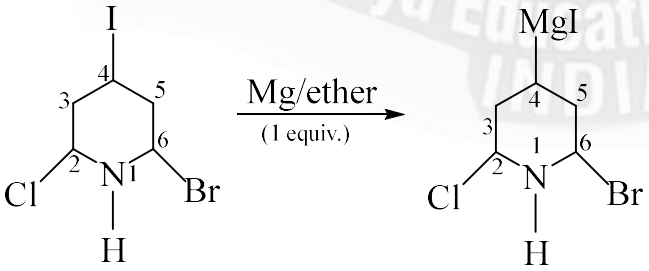


4-methyl pent-2-yne

$x = 8$

$\therefore \quad \frac{y}{x} = \frac{5}{8} = 0.625$

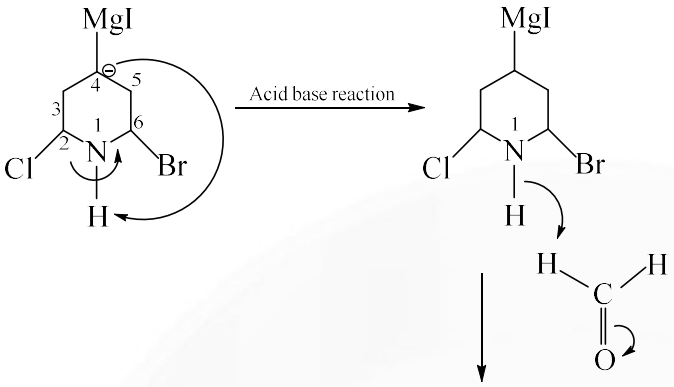
40.



[Reactivity order: $I > Br > Cl$]

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$\therefore\ p = 4$



$\therefore\quad q = 1$

Find value : $p^q = 4^1 = 4$

41. n-butane: $\overset{-3}{CH_3} - \overset{-2}{CH_2} - \overset{-2}{CH_2} - \overset{-3}{CH_3}$

iso-butane : $\begin{array}{l} \quad\quad\overset{-3}{CH_3} \\ \quad\quad| \\ \overset{-3}{CH_3}-\overset{-1}{CH}-\overset{-3}{CH_3} \end{array}$

n-tetraphosphane : $\overset{-2}{H_2}P-\overset{-1}{PH}-\overset{-1}{PH}-\overset{-2}{PH_2}$

Iso-tetraphosphane: $\begin{array}{l} \overset{-2}{PH_2}-\overset{0}{\overset{\cdot\cdot}{P}}-\overset{-2}{PH_2} \\ \quad\quad| \\ \quad\quad\underset{-2}{PH_2} \end{array}$

Total atom(s) in -2 oxidation state $2 + 0 + 2 + 3 = 7$

42. $V\ (Z = 23)$

$Y\ (Z = 39)$

$W\ (Z = 74)$

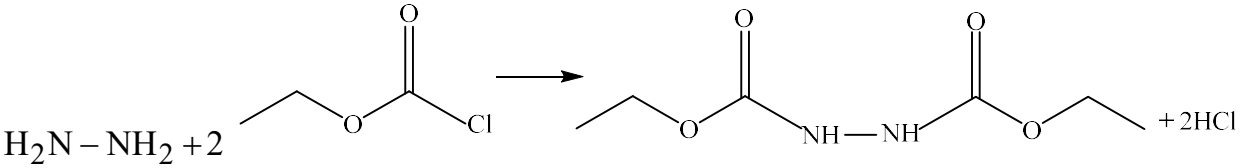
$U\ (Z = 92)$

Sum $= 23 + 39 + 74 + 92$

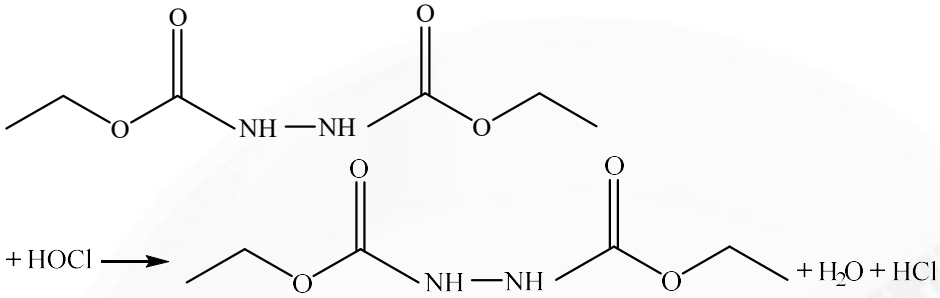
$= 228$

43. Step: 2

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Step: 2



Moles of HCl produced = 2+1 = 3

44. Factual

45. NCERT, Factual

	F_2	Cl_2	Br_2	I_2	–
Melting point/K	54.4	172.0	265.8	386.6	–
Boiling point/K	84.9	239.0	332.5	458.2	–
Density/g cm^{-3}	1.5 (85)[c]	1.66 (203)[c]	3.19(273)[c]	4.94(293)[d]	–
Distance X – X/pm	143	199	228	266	–
Bond dissociation enthalpy /(kJ mol^{-1})	158.8	242.6	192.8	151.1	–
$E^{\ominus}/V$[e]	2.87	1.36	1.09	0.54	–

$I-Cl \simeq 208$ kJ/mol

$F-F \simeq 158$ kJ/mol

46. Limonite is $Fe_2O_3 \cdot xH_2O$ and contains only ferric ions.

(b) $3Fe(s)+4H_2O(g) \xrightarrow{1073\,K} Fe_3O_4(s)+4H_2(g)$

(c) Fe_3O_4 has an inverse spinel structure where octahedral voids contain a mix of both Fe^{2+} & Fe^{3+} ions.

(d) Fe_3O_4 is ferrimagnetic

47. The given structure is Retinol (Vitamin A) & it is a fat-soluble vitamin, its deficiency leads to night blindness.

48. (a) $-I$ effect: $-CHO > -OCH_3 > -OH$

(b) moving $-OH$ to the ortho position decreases acidity due to strong intra-molecular H-bonding.

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(c) Replacing $-CHO$ with $-COOH$ group increases acidity as $-COOH$ itself will become the most acidic functional group in the molecule.

(d) meta – placement of $-OH$ removes resonance stabilization of the phenoxide ion by $-CHO$ thus lowering acidity

49. (a) Viscosity of liquids decreases as temperature rises because higher thermal energy increases molecular motion, allowing particles to overcome cohesive forces and slip past one another more easily.

(b) increase in branching creates spherical shape which decreases surface area, hence intermolecular forces, hence resistance to flow, hence viscosity

(c) Octanol is more Viscous than Octane due to intermolecular H-bonding.

(d) Propane-1, 2, 3-triol has more -OH groups for intermolecular H-bonding than ethanol.

50. $4PH_3(g) \rightarrow P_4(g) + 6H_2(g)$

(a) at $t = 0$, total pressure includes introduced PH_3 gas & the inert gas.

(b) at $t = 60s$, the flask contains inert gas, remaining PH_3 & produced P_4 & H_2.

(c) inert gas remains in the flask at $t = \infty$

(d) Since the reaction goes to completion, it must be zero order; the unit mentioned is inconsistent with expected order, so option(d) must be wrong

51. Balanced reaction is:

$$6Cu_3P + 11Cr_2O_7^{2-} + 124H^+ \longrightarrow 18Cu^{2+} + 6H_3PO_4 + 22Cr^{3+} + 53H_2O$$

52. (P) acid + metal = salt + $H_2(g)\uparrow$

(Q) CH_3MgBr + acidic 'H' = $CH_4(g)\uparrow$

(R) $\begin{matrix} CH_2-COOH \\ | \\ CH_2-COOH \end{matrix} \xrightarrow[\text{Kolbe's electrolysis}]{NaOH} \begin{matrix} CH_2 \\ \| \\ CH_2(g) \end{matrix}$

(S) $\begin{matrix} CH_2-COOH \\ | \\ CH_2-COOH \end{matrix} \xrightarrow[\text{decarboxylation}]{NaOH/CaO\ \Delta} \begin{matrix} CH_3 \\ | \\ CH_3 \end{matrix}$ ethane

53. - SCl_6 doesn't exist due to steric hindrance

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- PH_3 is a Drago's molecule (central atom doesn't undergo hybridisation when it is period-3 element onwards having at least one lone-pair & electronegativity of surrounding atom < 2.5)
- CCl_4 & PH_3 has exact no. of electron to form covalent bond(s) & satisfy octet rule.
- $AlCl_3$ is electron deficient and act as a Lewis acid.

54. I. Virgin Linseed Oil → P

Linseed oil is primarily composed of triglycerides containing fatty acids like alphalinolenic acid, linoleic acid, and oleic acid. Structure P is a large triglyceride molecule with long hydrocarbon chains (represented in red, blue, and green).

II. Diclofenac Diethylamine → Q

Diclofenac is a non-steroidal anti-inflammatory drug (NSAID). Structure Q clearly shows the dichlorinated phenyl ring, the secondary amine linker, and the phenylacetic acid group, paired with a diethylamine molecule (the salt form used in topical gels).

III. Methyl Salicylate → T

Methyl salicylate is an ester of salicylic acid. Structure T shows the benzene ring with a hydroxyl group $(-OH)$ and a methyl ester group $(-COOCH_3)$ at the ortho position.

Note: Structure R is actually Aspirin (acetylsalicylic acid), which is often a distractor in these problems.

IV. Menthol → S

Menthol is a cyclic terpene alcohol. Structure **S** is the characteristic cyclohexane ring with an isopropyl group and a methyl group, along with the hydroxyl $(-OH)$ group.

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